

A steady gain of 5% per year means the initial value of 1 goes up to 4.33 after 30 years. A steady gain of 10 percent per year means that initial value rises to 17.45 over the same time. The yearly rate has doubled. *But the end result after 30 years is over four times as much.* The longer the time period, the bigger the effect.

Of course, there is price volatility along the way with any investment portfolio. You won't get a perfectly smooth curve. But what matters for long-term investors is the end result. And the power of compounding still works its magic, even when there is price volatility along the way.

Most people don't understand compounding and exponential growth. This isn't because they are stupid or uneducated. It's just that it's not intuitive to most people. We're not wired to think in a nonlinear way. But it's increasingly essential to start to think this way if we are to truly understand the world we live in.

Mine's a Double!

If you know the rate at which something is growing, you can work out how long it will take to double in size (assuming the growth rate stays the same). The rate of growth is the percentage that it goes up each year. The number of years taken to double in size is called the "doubling time."

[Figure 3.4](#) summarizes the doubling times using a range of different annual growth rates.

Figure 3.4 Annual Growth Rates and Their Doubling Times

Annual growth rate (%)	Doubling time (years)
1	70.0
2	35.0
3	23.3
4	17.6
5	14.2
6	11.9
7	10.3
8	9.0
9	8.0
10	7.3
15	5.0
20	3.8
25	3.1

At lower rates of growth, small differences in the rate make a big difference to the doubling times. If your investments beat price inflation and taxes by 1%, they will